UNIT-4 INDUSTRIAL PHARMACY

ONESHOT - CRASH COURSE

B PHARMACY 5TH SEMESTER

IMPORTANT QUESTIONS

M.I.

1. Define and types of parenteral products. Write detailed note on:
   - Production of Parenteral products (procedure, facilities, control)
   - Quality control tests for parenteral products.

M.I.

Elaborate Lyophilized product (preparation of dry powder of lyophilization)

2 M.I.

Discuss formulation consideration/approaches and preparation of ophthalamic preparation (Eye drops, eye ointments and eye lotions) (M.I.)

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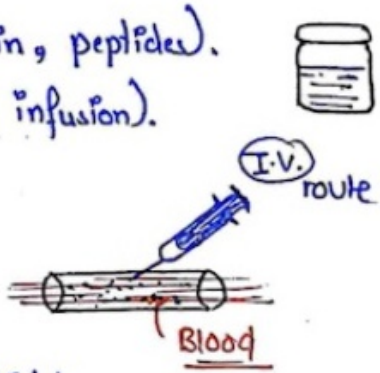
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Q1 Define and types of Parenteral Products. Write detailed note on:-

- Production of Parenteral products (procedure, facilities, controls).
- Quality control test for parenteral products.
- Elaborate Lyophilized product (prep" of dry powder by Lyophilization) + Prefarmulation factors

Parenteral Products

- Parenteral products are those sterile preparation which are given directly into the body through injection or infusion. These products bypass the gastrointestinal tract and therefore require very high standards of purity, sterility and safety.
- These are mainly used:-
  - When rapid drug action is required.
  - When patient cannot take medicine orally (unconscious, vomiting).
  - When drug is not effective orally (insulin, peptides).
  - For controlled and targeted delivery (IV infusion).



Advantages -

- Fast action - drug enters bloodstream immediately.
- Useful when oral route not possible - unconscious or vomiting patients.
- 100% bioavailability - complete dose is available.
- Suitable for drugs not absorbed orally - like insulin.
- Accurate dose and controlled delivery - especially IV infusion.

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Limitation -

- Painful - injections cause discomfort.
- Risk of infection - requires strict sterility.
- Needs trained person - cannot be self-administered safely.
- Costlier - special facilities and storage needed.
- Cannot be reversed - once injected, drug cannot be removed.

Types of Parenteral Products.

A. Based on volume
- Small volume Parenterals (SVP)
  Volume: ≤ 100 ml.
  Eg. Ampoules, Vials, Insulin injection.
- Large volume Parenterals (LVP)
  Volume: > 100 ml
  Eg. 500 ml Dextrose injection, 1L Normal saline.

B. Based on Route of Administration -
- Intravenous (IV) - directly into the vein.
- Intramuscular (IM) - into the muscle.
- Subcutaneous (SC) - under the skin.
- Intradermal (ID) - into the dermis.
- Intrathecal / Epidural - into spinal fluid.
- Intra-articular - into joint space.

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C. Based on Physical Form -

- Aqueous solutions - Eg. Adrenaline injection.
- Oily solutions - Eg. Vitamin A injection.
- Suspensions - Eg. Penicillin G procaine.
- Emulsions - Eg. Fat emulsions.
- Dry powders for reconstitution - Eg. Ceftriaxone dry powder.
- Lyophilized products - freeze dried sterile powders.

D. Based on Container Type -

- Ampoules.
- Vials.
- Infusion bottles / bags.
- Prefilled syringes / cartridges.

Characteristics of Parenteral Products.

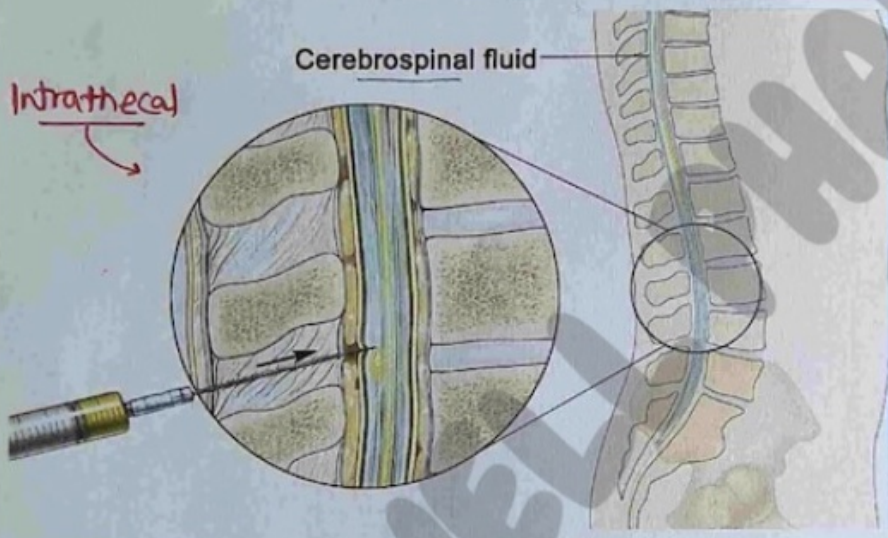
- Must be sterile (free from microorganisms)
- Must be pyrogen-free
- Should be free from particles
- Must have proper isotonicity.
- Should be stable during storage.
- Must be prepared under aseptic conditions.

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PARENTRAL PRODUCTS









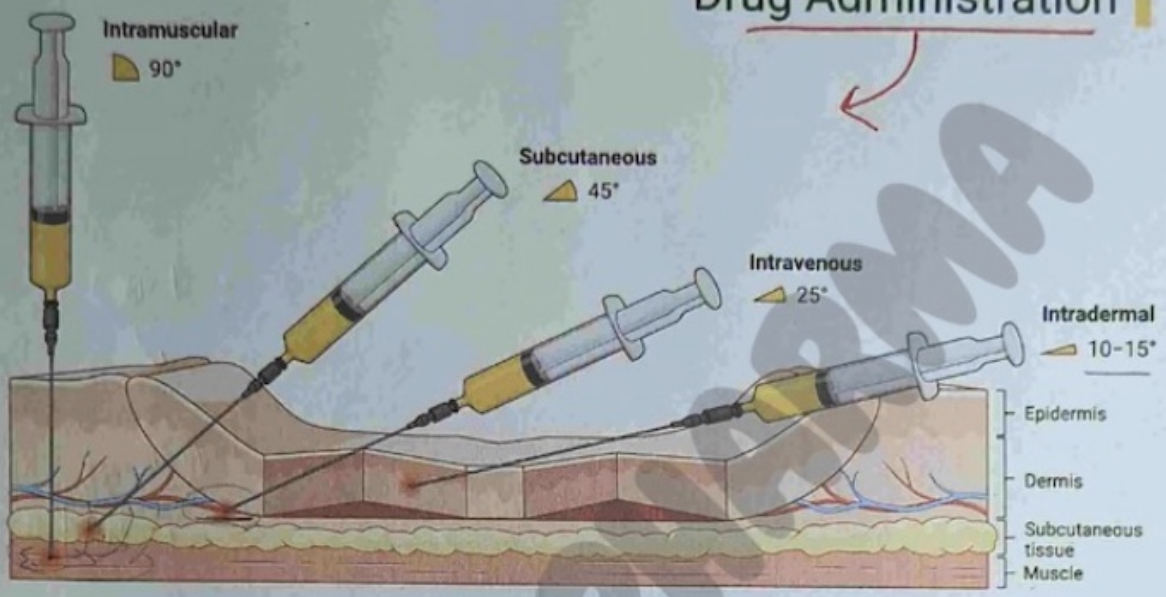


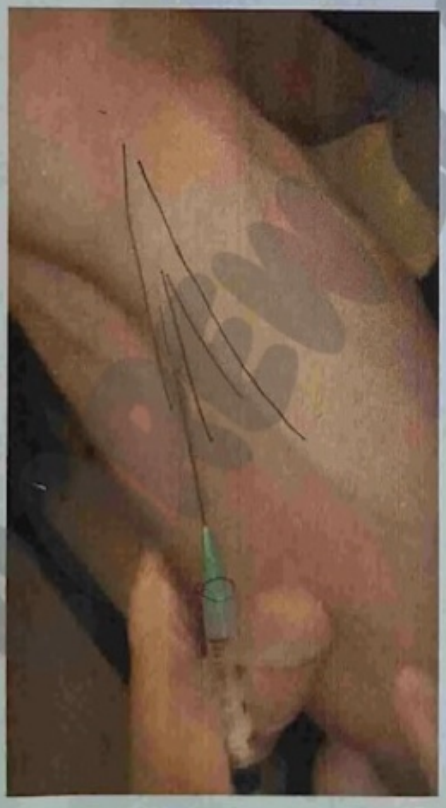






Parenteral Route of Drug Administration





Intra-venous (I.V.)



Laminar-air-flow

(LAF)

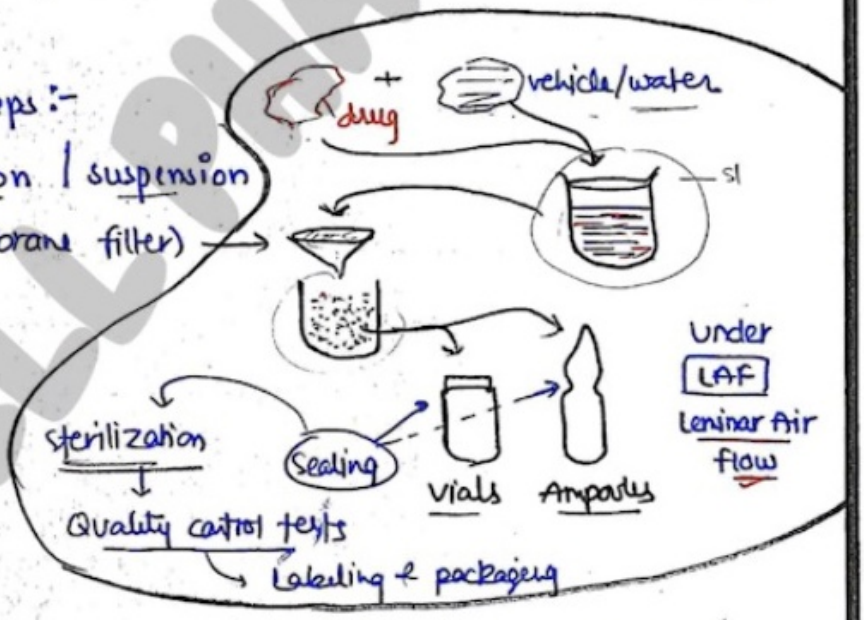
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MANUFACTURING/PRODUCTION OF PARENTERALS

Production of Parenteral Products

Parenteral products require very strict manufacturing conditions because they enter the blood body directly through injection.

- Their production must ensure complete sterility, removal of particle, correct composition and protection from contamination
- Parenteral manufacturing follows a fixed sequence of carefully controlled steps.
- It includes several steps :-
  - Preparation of solution / suspension
  - Filtration (sterile membrane filter)
  - Filling
  - Sealing
  - Sterilization
  - Inspection
  - Labeling & Packaging.



1. Preparation of solution / Suspension - The drug is dissolved or uniformly mixed in a suitable vehicle like water for injection or an oil. The pH and isotonicity are adjusted to make the injection safe for tissues. Undissolved particles are removed to keep the preparation clear.
   Eg. Adrenaline is prepared as a clear aqueous solution.

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2. Filtration - The prepared solution is passed through a sterile membrane filter (usually 0.22 μm). This removes microorganisms and very fine particles.. Filtration is essential especially when the product cannot tolerate heat sterilization. (eg. Heat sensitive antibiotics are sterilized only by filtration.

3. Filling - The sterile solution is transferred into ampoules, vials or infusion bottles under a Laminar Air flow (LAf) environment. This ensures that no microorganisms enter the product during filling. The volume filled must be accurate for each dose.

4. Sealing - The containers are immediately sealed to prevent contamination. Ampoules are sealed by melting the glass neck, while vials are closed using sterilized rubber stoppers and aluminum caps. Proper sealing maintains sterility during storage and transport.

5. Sterilization - After sealing, the product is sterilized either by heat (autoclaving) or by aseptic filtration, depending on the drug's stability. Heat-stable products undergo terminal sterilization, while heat-sensitive drugs are filled aseptically after filtration.

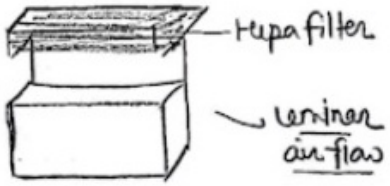
6. Inspection - Every filled container is checked visually for particles, cracks, leaks, discoloration or any defect. Inspection ensures that only safe and clear injections reach the patient. Quality control tests

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7. Labeling and Packaging - Finally, containers are labeled with product name, route (IV/IM/SC), batch number, expiry date, and storage conditions. Light-sensitive injections are packed in amber container to prevent degradation.

Product Facilities For Parenteral Products

- Parenteral products need special controlled facilities to avoid contamination. These areas maintain clean air, pure water and aseptic conditions during manufacturing.



Facilities

1. Clean Rooms - Controlled rooms designed to reduce particles and microbes.
   eg. ISO Class 5 area for filling.
2. Laminar Air flow (LAF) - Provides sterile, unidirectional air at working station.
   eg. Used during filtration and filling.
3. HEPA filters - Remove almost all airborne particles from incoming air.
   eg. used in clean room ceiling units.
4. Air Handling Unit (AHU) - Controls temperature, humidity and maintains positive pressure.
   eg. Prevents outside air from entering.



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5. Water for Injection (WFI) - Ultra-pure, pyrogen-free water used in preparation of injections.
   Eg. Used to dissolve drugs for (IV) injections.
6. Sterilization Area - Contains equipments for sterilizing materials and components.
   Eg. Autoclave for steam sterilization.
7. Aseptic Area - Highly clean zone used for critical operations like filling and sealing.
   Eg. Vial filling room.
8. Enviromental Monitoring Room - Used to test air, surfaces and personnel for microbial contamination.
   Eg. Air sample using settle plates.
9. Quarantine & Storage Area - Product stay here until all tests are passed
   Eg. Batches kept for sterility approval.

Production Controls for Parenteral Products.

During parenteral manufacturing, strict controls are applied at every stage to ensure sterility, safety and quality. These controls help prevent contamination and ensure that each batch meets official standards.

- It includes :—
- Enviromental Controls —
- Personnel controls —
- Sterility Assurance Controls :—
- Container - closure Integrity control —
- Documentation Control —

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- Environmental Controls - These include maintaining temperature, humidity, air pressure and monitoring microbial load in the manufacturing area.
  Eg. Measuring airborne microbes using settle plates.
- Personnel Controls - Ensures that workers follow strict grooming procedure and aseptic techniques. Only trained people are allowed in sterile zones.
  Eg. Use of sterile gloves, masks and gowns.
- In-Process controls - Quality checks performed during manufacturing such as pH, clarity, fill volume, particle check and filter integrity.
  Eg. Checking fill volume after every few vials.
- Sterility Assurance Controls - Ensures the sterilization processes are working correctly, including validation and routine checks.
  Eg. Biological indicators used in autoclave validation.
- Container-Closure Integrity Control - Ensures that ampoules, vials and bottles are sealed properly with no leakage or cracks.
  Eg. Dye ingress test to check sealing.
- Documentation Control - All manufacturing steps, cleaning steps and testing results are recorded to maintain traceability and GMP compliance.
  Eg. Batch manufacturing record.

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Quality Control Tests for Parenteral Products

Parenteral Products enter directly into the bloodstream, they must be completely (free) from microbes, particles and defects. for this reason, specific quality control tests are performed to ensure that every injection is sterile, safe, clear and contains the correct amount of drug. These tests are compulsory before releasing any parenteral batch.

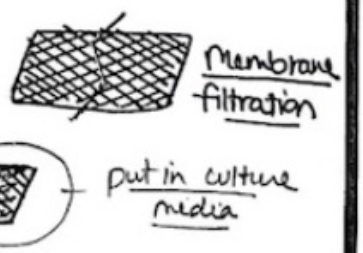
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1. Sterility Test - It confirms that the parenteral product is completely (free) from living microorganisms.
The test is performed by incubating the product in sterile culture media for (14) days and checking for microbial growth.

- Sterility testing uses two official methods:-

(A). Membrane filtration Method - In this method, the product is passed through a (sterile) membrane filter (0.4 μm).

- The filter traps any microorganisms present in the product
- After filtration, the membrane is placed in two culture media (FTM and SCDM) and incubated for 14 days.
- If (no) growth appears → the product is sterile.



Suitable for: heat sensitive drugs
- Aqueous solutions
- Alcohol-based solutions
- (Large volumes)

(B) Direct Inoculation Method - In this method, a small quantity of the product is added directly into the culture media (tubes).

- The tubes are then incubated and observed for microbial growth
  If the media remains clear → the product is sterile

Suitable for :-

- Oily injections
- Viscous products
- Products that cannot be filtered.



2. Pyrogen Test / Bacterial Endotoxin Test (BET).

- Pyrogens are fever-causing (substances), mainly (endotoxins) from Gram-negative bacteria.
- Even if a product is (sterile), pyrogens may still be present, so this test confirms that pyrogen level is safe.

Two methods :-

Bacterial Endotoxin Test (LAL) Test - Modern Method) -

Uses blood extract from horseshoe crab (Limulus Amebocyte Lysate).

- If endotoxin is present, the reagent forms a (gel) or turbidity.
- Very sensitive and fast.

Rabbit Test

- Product injected into rabbits.
- If body temperature (rises) above the limit → pyrogens present.
- Pyrogens-free product ensures no fever, chills or shock in patients.

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3. Leak Test / Container - Closure Integrity Test -

This test ensures that the container is sealed properly and no contamination can enter.

- Common Method - Dye Ingress Test.
  - Vials / ampoules are dipped in a colored dye solution (like methylene blue).
  - Vaccum is applied.
  - If dye enters the container → leakage present.

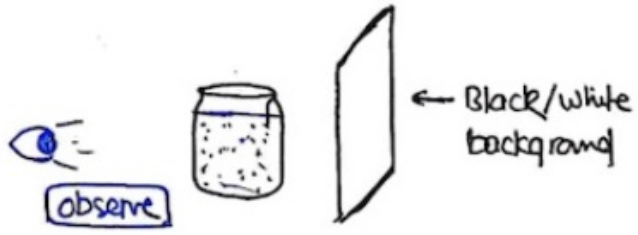


- Another Method - Vacuum leak test
  - Checking pressure changes.
- This test confirms that sealing is perfect and sterility and is maintained during storage.

4. Clarity Test - It ensures that the injection is clear, colorless and free from visible particles.

Method:-

- Hold the container against black and white background.
- Observe under diffused light.
- No haze, fibres, crystals or turbidity should be present.
- Clear solutions indicate:
  - Proper filtration.
  - No contamination.
  - No precipitation of drug.



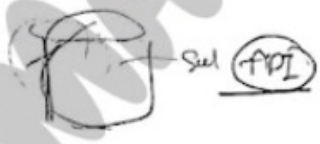
- This is essential because particles can block blood vessels.

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5. Assay And Content Uniformity Test -

Assay determines the actual amount of drug in the product. Content uniformity ensures each unit (each vial / ampoule) contains almost the same amount of drug.

Method -



- Chemical analysis (HPLC, UV spectrophotometry).
- Multiple containers are tested.
- Drug content should remain within acceptable limits (typically 90 - 110% depending on pharmacopoeia).
- Ensures patient receives correct and safe dose every time.

Lyophilized Products (Freeze - Dried Products).



- FORMULATION OF LYOPHILIZED PRODUCTS

Lyophilization, also called freeze drying, is a special technique used to convert heat-sensitive sterile solutions into a dry, stable powder,

- These products are later reconstituted with sterile water before injection. This method protects the drug from heat, moisture and degradation. and increases its shelf-life.
- Lyophilized products are sterile dry powders prepared by freezing a solution and then removing water directly as vapour under vaccum. They are mainly used for heat-sensitive drugs, antibiotics, vaccines and hormones.

Advantages:

- Suitable for heat sensitive drugs (cannot be autoclaved)
- Gives excellent stability during storage.

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- Produces a porous dry cake that dissolve quickly.
- Allows easy transport and storage.
- Extends shelf-life of the product.

Procedure

- Preparation of solution.
- Filling into vials.
- Freezing
- Primary Drying (sublimation)
- Secondary Drying (Desorption).
- Sealing and Packaging.

Step 1 :- Preparation of solution.

The drug is dissolved in a suitable sterile solvent (mostly water for injection).

- Stabilizers, buffers and bulking agents may be added.
- The solution must be clear, sterile and particle-free before moving to the next step.

Step 2 :- Filling into Vials

The sterile solution is filled into clean, sterile vials under aseptic conditions, usually inside Laminar Air flow units.

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Step 3:- Freezing.

The filled vials are placed on lyophilizer shelves and cooled to −40°C or lower.

- Water in the product becomes solid ice.
- Proper freezing forms a good internal structure, which creates the porous cake after drying.

Step 4:- Primary Drying (Sublimation).

This is the most important step.

- A strong vacuum is applied so that ice changes directly from solid to vapour without melting – this process is called sublimation, solid → vapours.
- Gentle heat is supplied to speed up sublimation.
- Most of the water (80–90%) is removed at this stage. A porous, sponge like cake starts forming inside the vial.

Step 5:- Secondary Drying (Desorption).

Any remaining bound water is removed by raising the temperature slightly (but below drug degradation point).

- This step reduces moistures to very low levels (0.5–3%)
- It ensures long-term stability of the dry powder.

Step 6:- Sealing and Packaging.

After drying is complete, vials are fully stoppered inside the lyophilizer under vaccum or nitrogen to prevent moisture entry. The vials are then sealed with aluminium caps and packed for storage.

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Characteristics

- Appears as a dry, uniform, porous cake.
- Reconstitutes quickly with diluent.
- Low residual moisture
- Stable during storage
- Free from collapse or shrinkage.

Advantages.

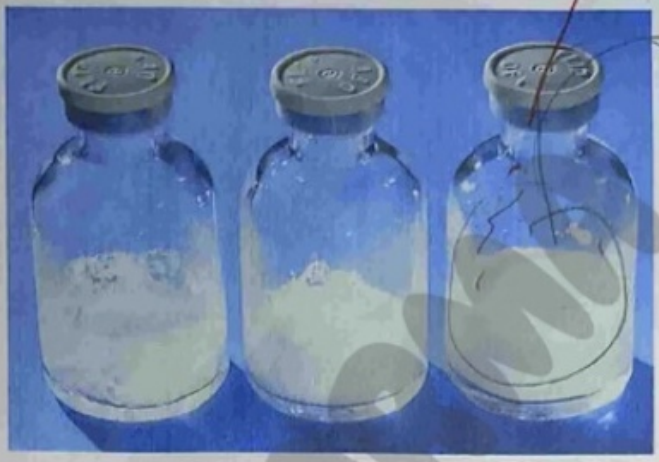
- Ideal for heat-sensitive and unstables drugs.
- Excellent stability and long shelf-life.
- Easy to transport
- Rapid reconstitution
- Maintains potency for a long time.

Limitations.

- Expensive equipment
- Long processing time
- Requires trained personnel.

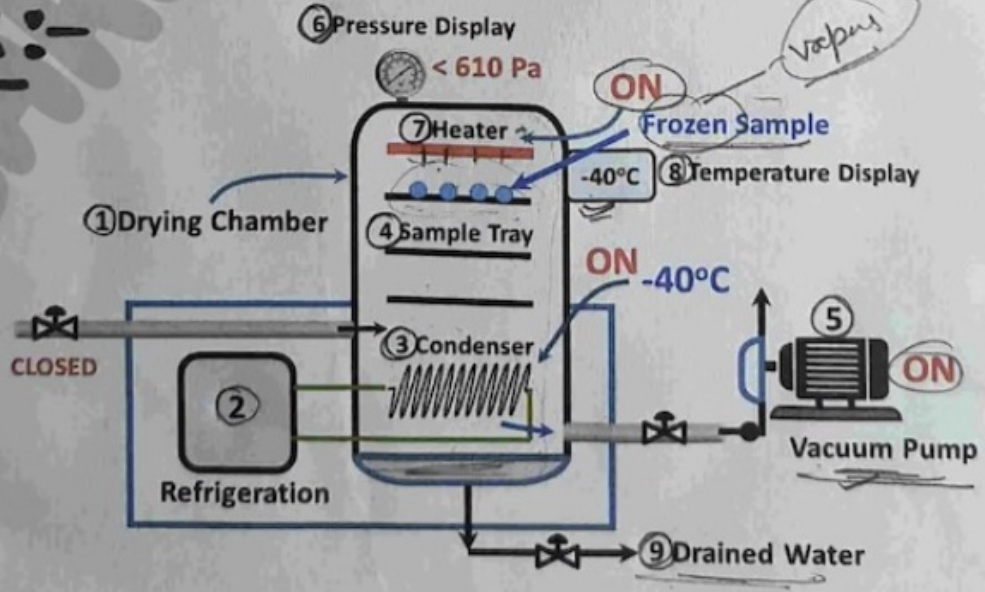
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LYOPHILIZED PRODUCTS + PROCESS





Steps :-



Preformulation Factors And Essential Requirements, Vehicles, Additives & Importance of Isotonicity

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Parenteral products are sterile injections administered directly into the body. Their formulation requires special attention to purity, stability, isotonicity and sterile. Before preparing any injectable product, certain preformulation factors and essential requirements must be considered along with suitable vehicles and additives.

(I) Pre-formulation Factors.

These are basic studies preformed befor formulation to ensure safety, stability and performance of the injection.

(i) Solubility - Drug must be soluble in suitable vehicle (mostly water)
Poorly soluble drugs need co-solvents or surfactants.

(ii) Stability - Drug should be stable against heat, light and pH.
If unstable → prepare as sterile powder.

(iii) pH of the Drug - Ideal pH of parenterals - 3-9 (comfortable range)

(iv) Compatibility with Excipients - Drugs must not react with buffers, preservatives or antioxidants.

(v) Tonicity Requirements - Drug solution should be isotonic to avoid pain or tissue damage.

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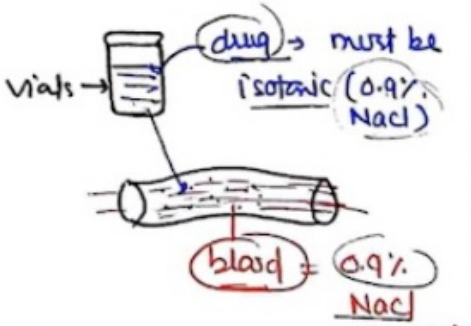
Essential Requirements of Parenteral Products.

(i) Sterility - Product must be free from microorganisms.
(ii) Pyrogen-free - No fever-producing endotoxins should be present.
(iii) Isotonicity - Must match osmotic pressure of body fluids.
(iv) Clarity and particle-free - Solution must be clear and free from visible particles.
(v) Suitable p^H - should not cause irritation or pain.
(vi) Stability - Product must remain stable during storage.
(vii) Safety and Non-toxicity - All ingredients must be non-irritant and safe.

Importance of Isotonicity.

Isotonicity ensures that the osmotic pressure of the parenteral product matches that of body fluids. It is important to:

- Prevents pain and irritation at injection site.
- Avoids hemolysis of RBCs (in hypotonic solution).
- Avoids shrinkage & crenation of RBCs (in hypertonic solution)
- Ensures patient comfort and safety.
- Prevents tissue damage.



How Isotonicity is Adjusted.

- using sodium chloride, dextrose or mannitol.
- Using sodium chloride equivalent method, freezing point depression method etc...

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Vehicles used in parenterals.

These are solvents used in preparation of parenteral products.

1. Aqueous vehicles - Water for injection (WFI) → primary solvent
   - Sterile water for injection → ready for use.
   - Bacteriostatic water → contains preservatives.

2. Non-Aqueous / oily vehicles - used for poorly soluble drugs or depot injection
   - sesame oil -
   - Cottonseed oil -
   - Peanut oil. -

   eg. Testosterone injection.

3. Co-solvents - Increase↑ solubility of poorly soluble drug.
   - propylene glycol -
   - Glycerol -
   - Polyethylene glycol. -

4. Emulsifying vehicles - used for fat emulsions

   eg. IV lipid emulsions.

Additives in Parenteral Products.

(i) Preservatives - used in multi-dose vials.

eg. Benzyl alcohol, phenol.

(ii) Antioxidants - Prevents oxidation

eg. Sodium metabisulfite, ascorbic acid.

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(iii) Buffers - Maintain suitable (pH) for stability and comfort
Eg. Phosphate buffer, acetate buffer.

(iv) Chelating Agents - Bind metal ions and prevents drug degradation.
Eg. EDTA.

(v) Stabilizer - Improve chemical and physical stability.
Eg Gelatin, albumin.

(vi) Tonicity Adjusting Agents - Ensure isotonicity.
Eg. Sodium chloride, dextrose.

(vii) Solubilizer / Surfactants - Increase solubility of (lipophilic) drugs.
Eg. Polysorbate 80 (Tween 80).

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Other topics :-

Formulation of Injections (Small volume Parenterals - SVPs)

- Injections are (sterile) liquid preparations (solution / suspension / emulsion) containing one or more drug substances. They are usually packed in ampoules or vials with volume less than (100) ml.

Components Used in Formulation.

(i) Drug - Active ingredients in pure, stable form. (eg). Diclofenac, Glucose, Ceftriaxone, NS, RL etc

(ii) Vehicle - Mostly water for injection (most common)
- Sometimes (oily) vehicles (sesame oil) for (slow) release.

(iii) pH Adjusting Agents - Buffers like phosphate or citrate to keep pH stable

(iv) Solubilizing Agents - Co-solvents to dissolve poorly soluble drug.
(eg). Propylene glycol.

(v) Antioxidants - prevents oxidation.
(eg). Sodium metabisulfite.

(vi) Preservatives (for multi-dose vials) - (eg) Benzyl alcohol. (to prevent microbial growth)

(vii) Tonicity Adjusters - Make injection isotonic
(eg). Sodium chloride.

(viii) Stabilizers - Protect (drug) from degradation.
(eg). EDTA.

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Special Considerations for formulating injections.

- Must be sterile
- Must be free from pyrogens.
- Solution must be clear, particle-free.
- Maintain stability during storage.
- Should be isotonic
- Suitable (pH) to avoid irritation.

Formulation of sterile Powders → Drug components

(i) Drug - Heat sensitive antibiotics, hormones, vaccines, proteins.

(ii) Bulking Agents - Provide structure to the dry cake

eg. Mannitol

(iii) Stabilizers - Protect drug during freezing and drying.

eg. Sugars like sucrose, trehalose.

(iv) Buffers - Maintain (pH) of solution

eg. Phosphate buffers.

(v) Cryoprotectants - Protect the drug from (freezing) damage.

eg. Glycine.

Special consideration

- Solution must be clear before (freeze) drying.
- Avoid (collapse) of cake
- Maintain low residual moisture.
- Ensures fast reconstitution with diluent.
- Use nitrogen filling if drug is oxygen-sensitive.

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Formulation of Large Volume Parenterals (LVPs).

LVPs are sterile solutions with a volume more than 100 ml (eg. 500 ml or 1L) They are used for IV infusion to supply fluids, electrolytes or nutrients.

Components used

(i) Solvent - Only water for injection (WFI)

(ii) Electrolytes - Sodium, potassium, calcium salts.
eg Sodium chlorides, potassium chlorides.

(iii) Carbohydrates - eg. Dextrose solution (5%, 10%)

(iv) Amino acids - For nutrition (TPN solutions) → Total Parenteral Nutrition

(v) Vitamins / Minerals - Added in special nutritional LVPs.

Special Consideration for LVPs.

- No preservatives allowed (large volume → toxicity risk).
- Must be isotonic
- Must be pyrogen-free
- Clear & particle free
- Packed in flexible bags or glass bottles.
- Sterilized by autoclaving.
- Must maintain chemical stability.

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Filling and Sealing of Ampoules and Vials.

- Ampoules and vials are commonly used containers for parenteral products. They must be filled and sealed under sterile conditions to prevent contamination. The filling and sealing process ensures correct volume, complete sterility, and protection of the drug throughout storage.

FOR AMPOULES

Steps of Ampoule filling.

(i) Washing and sterilization
(ii) Filling
(iii) Pre-heating
(iv) Sealing (flame sealing)
(v) Inspection.

(i) **Washing and sterilization** - Empty glass ampoules are washed to remove dust and particles. They are then sterilized using dry heat to ensure complete microbial removal.

(ii) **Filling** - Sterile solution is filled into ampoules using automatic filling machines under Laminar Air flow (LAf).

- Filling needles ensure correct volume.
- This step must be done aseptically.

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(iii) Pre-heating - Before sealing, the neck of the ampoule is gently pre-heated to make it soft.

- This makes sealing faster and prevents cracks.

(iv) Sealing (Flame sealing) - The ampoules neck is rotated in front of a gas flame (usually LPG + oxygen).

- The gass melts and forms a tight seal.
- This protects the sterile product from air and contamination.

(v) Inspection - Sealed ampoules are checked for visible particles, cracks, improper seals and correct volume.

- Only perfect ampoules are accepted.

FOR VIALS

Filling and Sealing of Vials.

- Washing and sterilization.
- Filling
- Partial stoppering.
- Lyophilization
- Final stoppering.
- Seating (Aluminium Crimping)
- Inspection.

- Washing and sterilization - Vials are washed inside-out using purified water and air jets. Then they are sterilized in a depyrogenative oven (dry heat).

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- Filling - Sterile liquid is filled into sterile vials under aseptic conditions.
  - Automatic filling machines ensure accurate volume.

- Partial Stoppering - Sterile rubber stoppers are placed on the vial but not fully pressed down.
  - This allows water vapour to escape during lyophilization.

- Lyophilization (If sterile powder is required) - Vials enter the lyophilizer for freezing, primary drying and secondary drying.
  - After drying, stoppers are fully pressed inside the lyophilizer chambers.
  (If liquid vial → this step is skipped)

- Final Stoppering - Rubber stoppers are fully inserted into the vial mouth.
  - This seals the vial internally.

- Sealing (Aluminium Crimping) - An aluminium cap is placed over the stopper and compressed tightly. This provides mechanical strength, prevents leakage and protects the stopper.

- Inspection - Vials are checked for cracks, leaks, particulate matter, correct sealing, and fill volume.

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Containers used in Parenterals.

Types of Containers

- Glass Ampoules
- Glass Vials
- Plastic infusion Bottles / Bags.
- Prefilled Syringes.

- **Glass Ampoules** → Made of Type I borosilicate glass.
  - Used for single dose injections.
  - Hermetically sealed → excellent protection.
    - 100% sealed (air-tight / leak proof. no air, bacteria, moisture, contaminant)
- **Glass Vials** → used for single-dose and multi-dose products.
  - Can store both liquids and sterile powders.
  - Closed with rubber stopper + aluminium cap.
- **Plastic Infusion Bottles / Bags** → used for large volume Parenterals (LVPs)
  - Lightweight, unbreakable, easy for handling.
  - Mostly made of PVC-free polymers (like PP or PE).
- **Prefilled Syringes** → Ready to inject dose.
  - Higher accuracy, reduced contamination risk.

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Closures used in Parenterals.

- Rubber stoppers (closures) → Made of (butyl) rubber.
  - Provide airtight and microbial seal.
  - Must be non-reactive, non-absorptive.
  - used on vials.
- Aluminium Caps (seals) →
  - Hold the rubber stopper firmly.
  - Prevent tampering and leakage
- Flip-off seals →
  - Plastic top for convenience
  - Easy access during injection preparation.

Labelling of Parenteral Products.

Label must include -

- Name of product -
- (Route) of Administration. (IV/IM/SC).
- Strength and volume.
- Storage conditions
- Batch number & expiry.
- Manufacturers details.
- Warning statement (eg. "Single dose only").
- Reconstitution instruction (for sterile powders).
- Ingredients (list) (excipients like preservatives).
- Light / Temperature protection instructions.

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Q.2 Discuss formulation consideration & preparation of ophthalamic preparations

Opthalmic Preparation

These are sterile medicines used in the eye. They must be free from particles, non-irritating and safe for the sensitive tissues of the eye.

- They are mainly used to treat infections, inflammation, dryness and for diagnostic purposes
  Eg. Eye drops, Eye ointments and Eye lotions (Eye wash).

Advantages -

- Provide direct action on the eye
- Give quick relief in local eye problems.
- Cause minimal systemic side effects.
- Available in multiple forms (drops, ointments, gels).

Limitation -

- Short contact time → washed away be tears.
- Need frequent dosing.
- Risk of contamination if not handled properly.
- Limited absorption due to corneal barrier.

(1) Eye Drops - These are clear, sterile liquid preparations meant for instillation into the eye.

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OPTHALMIC PREPARATION



Eye Drops

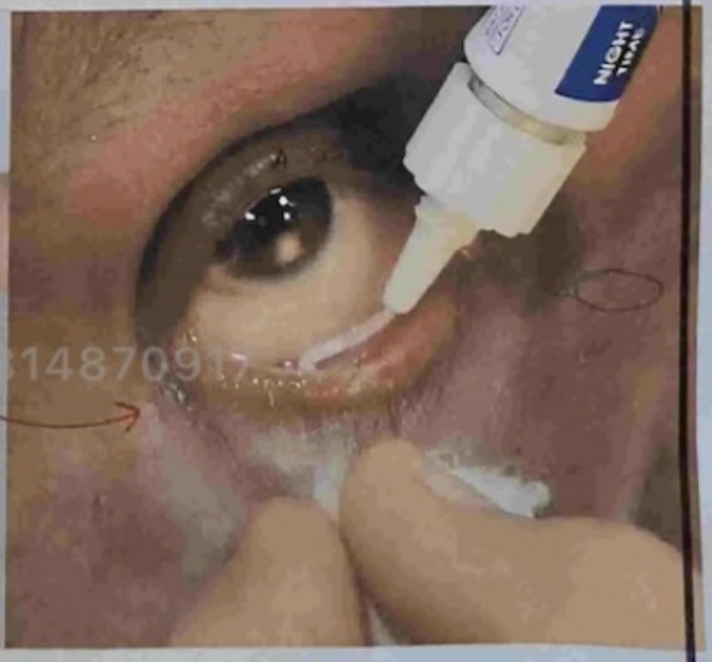




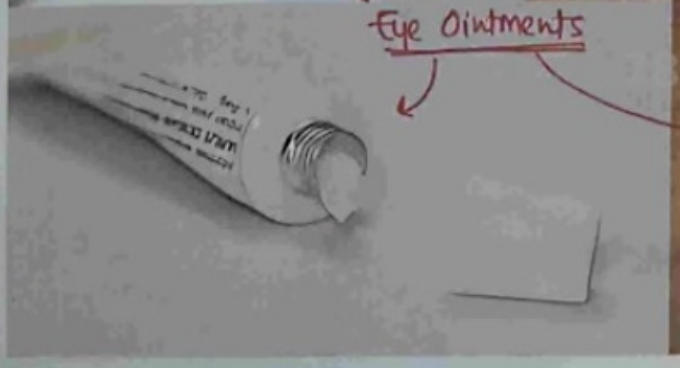


Eye lotion





Eye Ointments



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Properties -

- Must be clear and particle-free
- Must be sterile and isotonic.

Uses -

- used for infections, dryness, allergy, inflammation, glucoma.

Example - Ciprofloxacin eye drops, Artificial tears.

2. **Eye lotions** - These are sterile, large volume aqueous solutions used for washing or bathing the eye.

Properties -

- Must be sterile, clear and non-irritating.
- Preservatives-free (used immediately).

Uses -

- used to remove foreign bodies, dust, chemical and for general irrigation.

Example - Normal saline eye lotion (0.9% NaCl), Boric acid lotion.

3. **Eye ointments** - These are sterile semi-solid preparations meant for application inside the lower eyelid.

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Properties -

- Made with smooth, non-irritant base
- Drug must be finely micronized.

Uses -

- Used for prolonged drug action, infection, inflammation and nighttime therapy.

Example - Erythromycin eye ointment, Tetracycline eye ointment.

Formulation Consideration and Preparation of opthalmic Preparations

- Opthalmic preparations are sterile medicines meant for application to the eye, such as eye drops, eye lotions and eye ointments.
- They must be formulated carefully because the eye is very sensitive. These preparations must be sterile, non-irritating, clear, isotonic and chemically stable.

A. Formulation Consideration.

(i) Sterility - All opthalmic product must be sterile to avoid eye infections. Sterilization may be done by:

- Autoclaving (for heat-stable solutions)
- Sterile filtration (0.22 μm) - for eye drops only
- Dry heat - for ointment base only.

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(ii) Clarity and Particle-free - The solution should be clear and free from visible particles.

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- Applies mainly to eye drops and eye lotions.
- Ointments must have finely micronized drug so particles don't scratch the eye.

(iii) Isotonicity - Tonicity must match natural tears to avoid pain or irritation.

- Adjust using -
  - Sodium chloride
  - Dextrose

  (Required for eye drops & eye lotions; not needed for ointments.

(iv) pH Adjustment / Buffering - pH should be close to tear pH (≈ 7.4).

Buffer used - Phosphate buffer.

- Borate buffer

(v) Viscosity Adjustment - Higher viscocity increases contact time of the drug with eye surface.

- Viscosity enhancers:-

  Methylcellulose, HPMC

- Mainly used in eye drops (not in lotions, rarely in ointements).

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(vi) Preservatives - Used in multi-dose containers to prevent contamination.
Eg. Benzalkonium chloride (BAK), chlorobutanol
- [Not added to eye lotion because they are used immediately; not required in single-use tubes of ointments].

(vii) Antioxidants - Added for oxidation-sinsitive drugs.
Eg. Sodium metabisulfite, EDTA

(viii) Ointment Base Requirements - (only for Eye ointment) -
Base must be smooth, non-irritant.
- Common bases - white soft paraffin, mineral oil.
- Must melt slowly at body temperature to remain in eye.

(ix) Containers -
- Eye drops → LDPE plastic dropper bottles.
- Eye lotions → large sterile bottles (for washing).
- Eye ointments → collapsible metal/plastic tubes.

Method of Preparation of opthalmic Preparations — Eye Drops

1. Preparation of solution (Dissolving the Drug) - To make a clear solution without particles.
- First, measured amount of drug is dissolved in sterile water for injection (WFI)
- If drug dissolves slowly → gentle stirring or warming allowed.
- If drug is poorly soluble → co-solvent (Eg. propylene glycol) is added.

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2. Adjusting pH using Buffers - It ensures comfort and drug stability
   - Eyes are very sensitive to pH changes
   - If solution is too acidic or too alkaline → burning sensation.
   - So, phosphate or borate buffer is added to maintain pH around 7-7.4.

3. Adjusting Isotonicity - To make solution comfortable for eye.
   - Teardrop tonicity ≈ 0.9% NaCl.
   - If solution is not isotonic → intense irritation & tearing.
   - Tonicity adjusters used:
   - Sodium chloride (most common)
   - Dextrose.

4. Adding Other Excipients - Depending on the drug, following excipients may be added:
   - Preservatives → for multi-dose bottles.
     (eg. Benzalkonium chloride.
   - Antioxidants → For oxidation-sensitive drugs.
     (eg. Sodium metabisulfite.
   - Viscosity agents → To increase retention
     (eg. HPMC, methylcellulose.

5. Filtration (Sterile filtration) - solution is passed through 0.22 μm membrane filter.
   - This removes microorganisms + particles.
   - It is the Best method for sterilization eye drops.

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6. Aseptic Filling - Sterile solution is filled into sterile LDPE plastic dropper bottles.
   - Filling is done inside Laminar Air flow (LAF) to keep enviroment sterile.

7. Sealing & Labelling - Bottles are sealed immediately.
   - Lables includes - drug name, strength, route (Eye use), storage, batch no., expiry.

8. Final Inspection - Bottles checked for clarity, leaks, scratches, correct fill volume.

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Preparation of Eye lotion

1. Dissolving Ingredients - Drug (if present) or cleansing ingredients are dissolved in Purified water.

2. Adjust pH & Isotonicity - Same as eye drops → pH near 7
   - Sodium chloride added for isotonicity.

3. Filtration for clarity - Must be particle-free, because lotions wash the entire eye surface.

4. Sterilization (Autoclaving) - Eye lotions are heat-stable, so sterilized by steam autoclave at 121°C.

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5. Filling - Filled aseptically into large sterile bottles.

6. Labelling - Clearly labelled "for Eye irrigation Only".
   - No preservatives → must be used immediately.

Preparation of Eye ointments.

1. Sterilization of ointment Base - To Destroy microbes in the base
   - Base like white soft paraffin or mineral oil is sterilized by dry heat.
   - Because water / steam cannot sterlize oily bases.

2. Sterilization of Drug Powder - To ensure comfort + sterility.
   - Drug powder is sterilized separately:
   - Either dry heat
   - Or gamma radiation.
   - Drug must be finely micronized so it doesn't scratch the eye.

3. Incorporation of Drug - Sterile drug powder is mixed into melted warm sterile base.
   - Mixing must be uniform, no lumps or aggregates.

4. Homogenization - The ointment is passed through an ointment mill or roller mill.
   - This ensures smoothness ↔ Very important for ophthalmic.

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5. Aseptic filling - warm ointment is filled into sterile collapsible tubes inside LAF.

6. Sealing & Labelling - Tubes sealed and labelled: drug name, strength, "for ophthalmic use", batch no, expiry.

Containers used & Labelling of ophthalmic Preparation.

Ophthalmic preparations need special containers to maintain sterility, clarity and safety.

- Common containers are:

(i) Plastic Dropper Bottle (LDPE) - for eye Drops.

- Most common used Squeezable, lightweight, unbreakable
- Provide controlled drop size
- Compatible for sterilization

(ii) Plastic or Glass Bottles - for Eye lotion

- Large sterile bottles (100 - 500 ml)
- Used for eye washing / irrigation.
- Generally preservatives-free (used immediately).

(iii) Collapsible Metal / Plastic Tubes - for Eye ointments.

- Made of aluminium or plastic
- Protects ointments from contamination
- Small openings to prevent excess flow.

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(iv) Single - dose Sterile Units -

- Used for preservatives - free formulations.
- Single - use plastic vials.
- Prevent contamination completely.

Labelling Requirements for Ophthalmic Preparations

- Ophthalmic preparations require strict and clear labelling for safe use because eye is a sensitive organ.
- Label must include:
  - Name of Preparation.
    eg. Ciprofloxacin Eye Drops.
  - Strength & Volume
    eg. 0.3 % w/v, 5 ml.
  - Route of administration.
    "for ophthalmic use only" must be written clearly.
  - Storage conditions
  - Store in cool, dry place.
  - Protect from light (if needed).
  - Preservative used (if any)
  - Batch number & expiry date
  - Sterility statement.
    eg. "Sterility until opened".
  - Shake well before use. (for suspension only).
  - Reconstituted instructions (if powders for eye drops).
  - Caution notes ("Do not touch dropper tip to any surface")
    - Use within 28 days after opening (usually 28 days).

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Quality Control Tests for Ophthalmic Preparation.

These are sterile, clear, isotonic and safe because they come in direct contact with the eye. Quality control tests ensure that the product is free from contamination, irritation and has the correct drug strength and sealing.

Ophthalmic preparation

Method Used.

- Sterility test
- Clarity / Particulate matter test
- pH test.
- Isotonicity test.
- Assay / content uniformity.
- Leakage test

(i) Sterility test – Ensures the product is free from microorganisms.

- Done by membrane filtration or direct inoculation.
- Incubated for 14 days → no growth – sterile.

Eg. Testing sterility of Ciprofloxacin Eye Drops before release.

(ii) Clarity / Particulate Matter Test – (for eye drops & lotions).

- Ensure solution is clear and particle-free.
- Observed against black & white background.

Eg. Checking clarity of Artificial Tear Eye Drops. →



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(iii) pH Test - Ensures (pH) is close to tear pH (~7.4) to avoid burning or irritation.

- Measured using a (pH) meter.

Eg. pH check for Moxifloxacin Eye Drops.

(iv) Isotonicity Test - Ensures solution's osmotic pressure matches natural tears. 0.9% w/v NaCl

- Measured by (osmolality) testing
- Non-isotonic → stining & excessive tearing.

Eg. Ensuring (0.9%) NaCl Eye lotion is isotonic.

(v) Assay / Content Uniformity - Confirms correct drug amount is present in each container.

- Tested by (HPLC) or spectrophotometry.

Eg. Assay of (0.3%) Gentamicin Eye Drops.

(vi) Leakage / Container - Closure Integrity Test -

- Checks whether the bottle or tube is properly (sealed) and has no leakage.

Eg. Squeezing the eye (drop) plastic bottle to check if any solution leaks out.

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